

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I D.Raghava(192525304) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D.Raghava

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

ANSWERS

1)A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

SOL) During seasonal sales, an e-commerce company may receive a sudden increase in website traffic. If the server cannot handle the load, the website may become slow or unavailable. Cloud **auto-scaling** and **load balancing** help manage this situation efficiently.

Cloud Auto-Scaling:

Auto-scaling automatically increases or decreases the number of servers based on traffic demand. During high traffic, new virtual machines or containers are added automatically. When traffic decreases, the extra resources are removed, reducing unnecessary costs.

Load Balancing:

A load balancer distributes incoming user requests evenly across all available servers. This prevents any single server from becoming overloaded and ensures all servers are utilized efficiently.

Reducing Downtime and Improving User Experience:

- Prevents server crashes during heavy traffic.
- Maintains fast page loading and smooth checkout.
- Ensures high availability by redirecting requests if one server fails.

Feature	Benefit
Auto Scaling	Adds servers automatically
Load Balancer	Shares traffic equally
Monitoring	Detects traffic spikes
Cloud	High availability

Role of Monitoring Tools:

Monitoring tools continuously track CPU usage, memory, network traffic, response time, and server health. When predefined thresholds are reached, they trigger auto-scaling and alert administrators about any issues. They also help identify performance bottlenecks and maintain system reliability.

Example:

- AWS EC2 Auto Scaling
- Elastic Load Balancer (ELB)
- Amazon CloudWatch

OR

- Microsoft Azure VM Scale Sets
- Azure Load Balancer
- Azure Monitor

OR

- Google Cloud Load Balancer
- Cloud Monitoring

Real-World E-Commerce Scenario:

During events such as **Amazon Great Indian Festival** or **Flipkart Big Billion Days**, millions of customers access the websites simultaneously. Auto-scaling launches additional cloud servers to handle the increased demand, while load balancers distribute customer requests evenly. Monitoring tools detect traffic spikes and automatically initiate scaling, ensuring the websites remain available, and responsive throughout the sale.

Conclusion:

Cloud auto-scaling, load balancing, and continuous monitoring work together to provide high availability, minimize downtime, optimize resource usage, and deliver a smooth shopping experience even during peak traffic periods.

2) An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Sol) Organizations generate large amounts of backup data that must be stored safely and be available when needed. **Cloud storage** provides a secure, reliable, and cost-effective solution for storing backups.

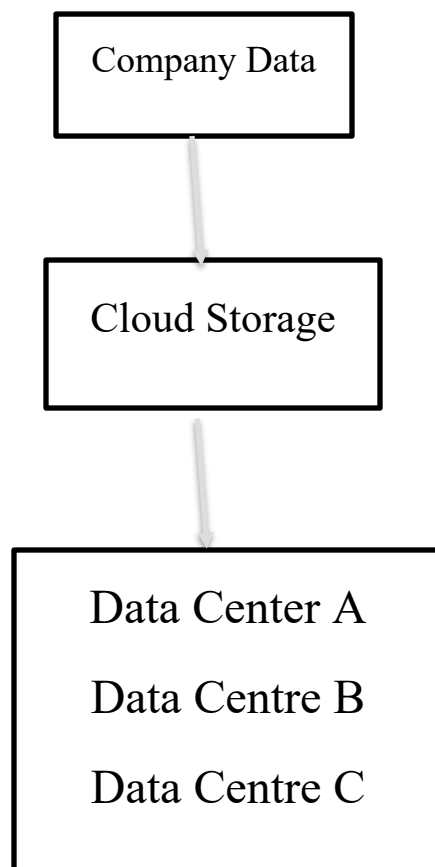
Cloud Storage Durability and Data Redundancy:

- Cloud providers store multiple copies of data across different servers and data centers.
- If one server or data center fails, the backup can be retrieved from another copy.
- This ensures high durability and prevents data loss due to hardware failures or disasters.

Encryption and Access Control:

- **Encryption** protects data by converting it into an unreadable format during transmission and while stored in the cloud. Only authorized users with the correct encryption key can access the data.
- **Access control** uses user authentication, passwords, role-based access control (RBAC), and multi-factor authentication (MFA) to ensure that only authorized users can view or modify backup data.

Cloud Backup Diagram:



Cost Advantages over Traditional Storage:

- No need to purchase and maintain expensive physical storage hardware.
- Pay only for the storage space actually used (pay-as-you-go model).
- Reduced maintenance, electricity, cooling, and hardware replacement costs.
- Storage capacity can be increased or decreased easily without buying new equipment.

Disaster Recovery Use Case: A company stores daily backups of its customer database in the cloud. If its office servers are damaged by a flood, fire, or cyberattack such as ransomware, the organization can quickly restore the latest backup from the cloud and resume business operations with minimal downtime and data loss

Cost Comparison Table:

Traditional Storage	Cloud Storage
Buy Hardware	Pay as You Use
High Maintenance	Low Maintenance
Limited Storage	Easily Scalable
Manual Backup	Automatic Backup

Security Technologies:

- AES-256 Encryption
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- SSL/TLS Secure Transmission

Conclusion:

Cloud storage offers high durability through data redundancy, strong security through encryption and access control, lower operational costs, and reliable disaster recovery, making it an ideal solution for modern backup and recovery systems.

3) A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Sol) A mobile application deployed globally must provide fast and reliable access to users in different countries. **Content Delivery Networks (CDNs), edge servers, and cloud infrastructure** help improve performance, reduce latency, and ensure high availability.

1. How CDNs Improve Performance and Reduce Latency

A **Content Delivery Network (CDN)** is a network of distributed servers located in multiple geographic regions. Instead of sending data from a single central server, the CDN delivers content from the server closest to the user.

Benefits:

- Reduces network latency.
- Faster loading of images, videos, and application content.
- Reduces bandwidth usage on the main server.
- Handles millions of users simultaneously.
- Improves application speed and reliability.

2. How Edge Servers Enhance User Experience:

Edge servers are located near end users and store frequently accessed content.

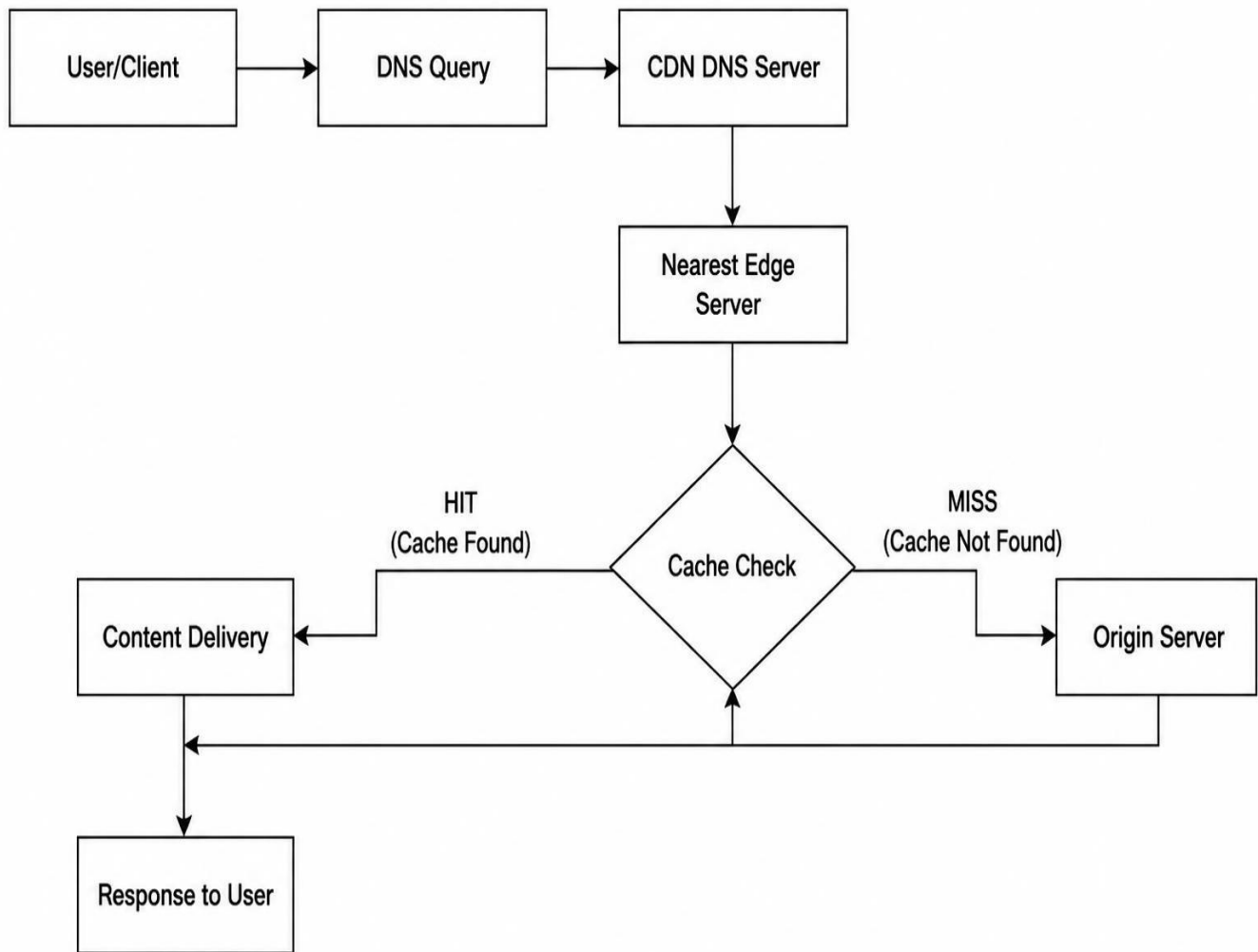
When a user requests data:

- The request is sent to the nearest edge server.
- The edge server delivers the content quickly.
- If the content is unavailable, it retrieves it from the origin server and caches it.

Advantages:

- Faster response time.
- Lower latency.
- Reduced buffering.
- Smooth user experience.
- Lower workload on the central server.

CDN Architecture:



3. Cloud Providers Support Global Availability:

Cloud providers maintain multiple **data centers** across different countries.

They provide:

- Global data centers.
- Automatic traffic routing.
- Load balancing.
- Auto-scaling.
- Disaster recovery.
- High availability (24×7 uptime).

If one region becomes unavailable, traffic is automatically redirected to another healthy region, ensuring uninterrupted service.

4. Real-Time Streaming / Gaming Example

Consider **Netflix** or an online multiplayer game like **PUBG Mobile**.

- A user in **India** receives video/game data from the nearest CDN edge server in India.
- A user in **Europe** connects to a nearby European edge server.
- The cloud automatically balances traffic across multiple regions.
- During peak hours, additional servers are launched automatically.
- This reduces buffering, lag, and game latency while providing a smooth viewing or gaming experience.

Conclusion

CDNs, edge servers, and cloud services work together to reduce latency, improve performance, ensure global availability, and provide a seamless experience for users worldwide.

4) A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Sol) Financial institutions handle highly sensitive customer and transaction data. Therefore, they must follow strict **regulatory compliance, security standards, and data protection policies** when using cloud services.

1. Cloud Providers Support Regulatory Compliance:

Cloud providers offer built-in security controls and certifications that help organizations comply with government and industry regulations.

They support compliance by:

- Following international security standards (ISO 27001, PCI-DSS, SOC 2).
- Providing regular security audits.
- Maintaining compliance reports and documentation.

- Enforcing identity and access management (IAM).
- Offering backup and disaster recovery services.

This helps financial institutions meet legal and regulatory requirements.

2. Role of Logging, Monitoring, and Encryption:

Logging

- Records every user activity and system event.
- Tracks login attempts, transactions, and file access.
- Helps during audits and investigations.

Monitoring

- Continuously monitors cloud resources.
- Detects unusual activities, cyberattacks, and unauthorized access.
- Sends alerts to administrators for quick response.

Encryption

- Encrypts data both **at rest** and **in transit**.
- Protects customer account details and financial records.
- Only authorized users with encryption keys can access the data.

3. Ensuring Data Sovereignty:

Data sovereignty means data must remain within the country's legal boundaries.

A company can ensure this by:

- Selecting cloud data centers located within the required country.
- Configuring data residency policies.
- Restricting cross-border data transfers.
- Following local government regulations.
- Regularly auditing data storage locations.

This ensures compliance with national data protection laws

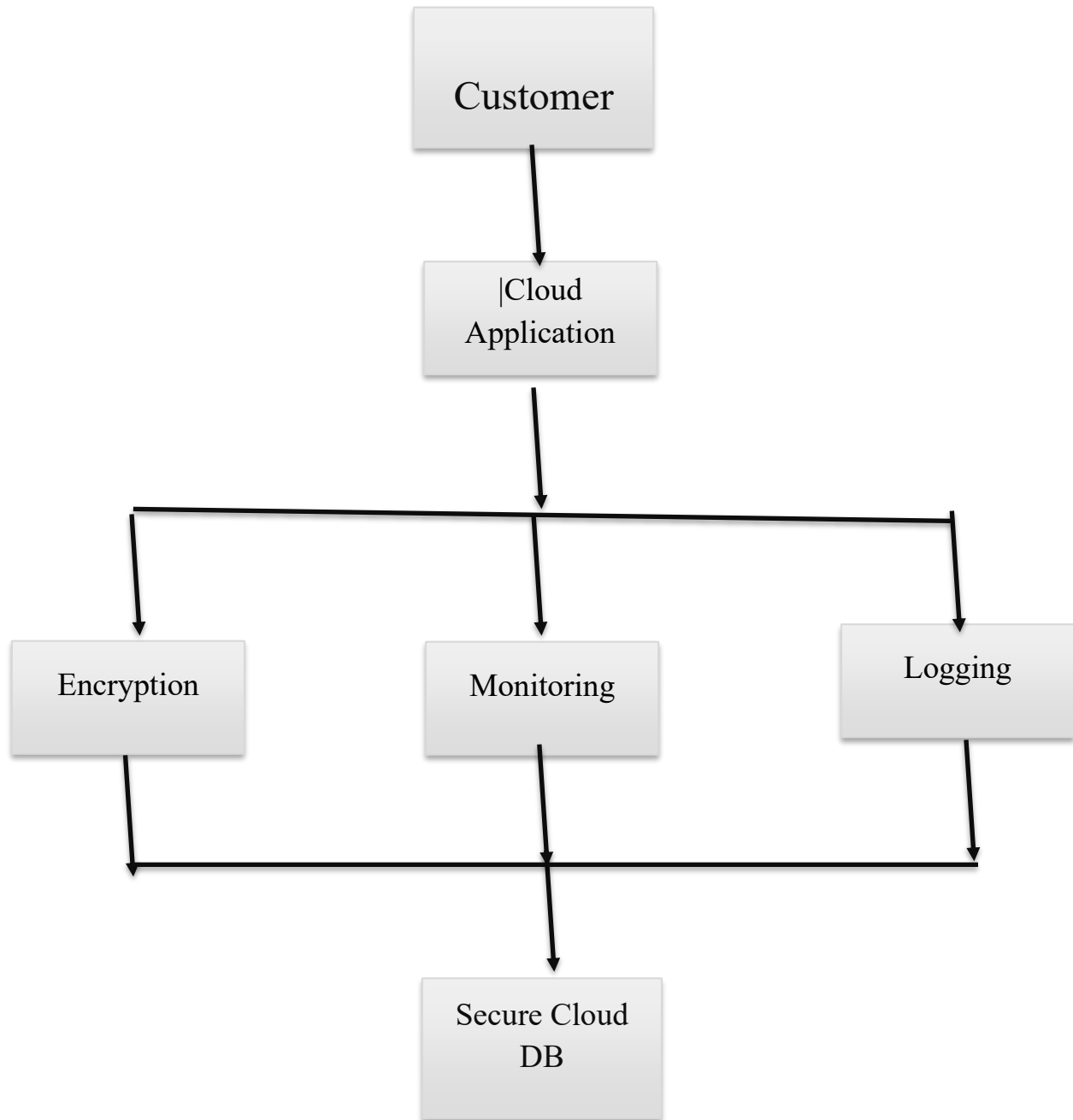
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4. Example – Sensitive Financial Transactions:

Consider an online banking system where customers transfer money between accounts.

- Every transaction is encrypted.
- All login attempts and money transfers are logged.
- Monitoring tools detect suspicious activities such as unusual login locations or large transactions.
- Customer financial data is stored in cloud data centers located within the country to satisfy data sovereignty requirements.
- If suspicious activity is detected, administrators receive immediate alerts and can take corrective action.

Security Flowchart:



Conclusion:

Cloud providers help financial institutions achieve **regulatory compliance**, **secure data through encryption**, **track activities using logging**, **detect threats with monitoring**, and **meet data sovereignty requirements**, ensuring secure and reliable financial services.

5) A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Sol)A development team can use **containerization** to package an application along with its dependencies into lightweight containers. Containers provide faster deployment, efficient resource utilization, and consistent performance across different environments.

1. Containers vs Virtual Machines

Containers	Virtual Machines
Share the host operating system	Each VM has its own operating system
Lightweight	Heavyweight
Fast startup (seconds)	Slow startup (minutes)
Low resource usage	High CPU and memory usage
Higher performance	Comparatively slower
Suitable for microservices	Suitable for running multiple OSs

Difference:

- Containers are lightweight because they share the host OS kernel.
- Virtual Machines require a complete guest operating system, consuming more CPU, RAM, and storage.
- Containers start much faster and allow more applications to run on the same hardware.

2. Kubernetes (Container Orchestration)

Kubernetes automates the deployment, management, and scaling of containers.

It simplifies deployment by:

- Automatically deploying containers.
- Scaling applications up or down based on demand.

- Restarting failed containers automatically.
- Load balancing traffic among containers.
- Performing rolling updates with minimal downtime.
- Managing container health and availability.

This reduces manual effort and improves application reliability.

3. Portability Across Cloud Environments

Containers package the application and all required libraries together.

Therefore, the same container can run on:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)
- Private Cloud
- Local Computer

This eliminates compatibility issues and supports "**Build Once, Run Anywhere.**"

4. Example – Microservices-Based Application

Consider an **online food delivery application (Swiggy/Zomato)**.

Each service runs in a separate container:

- User Service
- Login Service
- Restaurant Service
- Order Service
- Payment Service
- Notification Service
- Delivery Tracking Service

If the **Order Service** receives heavy traffic, Kubernetes automatically creates additional containers only for that service without affecting the others. This improves scalability, availability, and performance.

Conclusion

Containers provide lightweight, fast, and portable application deployment. Kubernetes automates deployment, scaling, and recovery, while microservices improve flexibility and maintainability, making containerization ideal for modern cloud applications.